

Prof. Karl Friston ~ Applied Active Inference

Symposium pt. 1 (Education)

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hello and welcome to the active inference lab to our first ever applied active inference symposium today it's june 21st 2021 and we're very honored to be here with professor carl fristen and many of our lab participants just as a way of quick introduction the active inference lab is a non-profit organization that is a participatory open science laboratory and we're working to curate and develop applications related to the active inference framework something that hopefully we'll be going into a lot more in detail today and this is a screenshot of our website as far as the overview of this symposium there are three organizational units in the lab edu education dot coms communication and dot tools and each of these units are going to facilitate a 45 minute or so session and we'll have a short break in between sessions so in our weekly meetings over the past weeks for each organizational unit we've been developing questions and getting excited about things that we wanted to talk to you about as far as a few overarching themes that were kind of spoken to um really through the whole journey of our lab but also across organizational units the first theme is applying active inference across systems again something that will come up probably in all sections the idea of research debt the idea that we don't want to be developing um research frameworks that have a huge burden on those who are learning and applying and that especially early in the formalization of frameworks it's extremely valuable to increase the accessibility so that we don't end up with major headaches and incompatibilities later on collective intelligence and the ways in which that is manifest across different systems transdisciplinary teams projects and communities which are kind of like nested levels of organization but transdisciplinarity is something that is necessary for the type of work that we're all interested in and also just modern challenges and opportunities for research and all that that means related to online and everything else and of course anything else that you have tumbling around and wanted to bring to the table thematically so there we are with our sort of lab overview and introduction let's go to our first organizational unit.edu the goal of edu is to scaffold and create a participatory and dynamic

active inference body of knowledge which we'll talk more about in a second and our progress and actions this year have been to release a terms list v1 which benefited greatly from your feedback and also we're now updating the terms list to version 2 which now includes five complete language translations and many references and citations for the terms the way that we're approaching the development of the terms is by using approaches that place ontology and progressively more formalized versions of ontologies as kind of the backbone of an educational body of knowledge so we started on the left side here with a terms list in the first quarter of 2021 and the ontology working group is like a train that's pushing to the right as they're learning ontology by doing and developing progressively stronger and stronger ways of relating the terms and the concepts that are essential for understanding active inference and this will help us develop principled educational material that's also able to be translated rapidly alex do you want to give a quick thought on where knowledge engineering comes into play yeah thanks um at this slide we are showing this work with anthology with system engineering approach which we are also using in the lab and considering possible deliverables of working on educational materials and creating them we should have at some point of time textbooks and educational courses and actually maybe this lab is started from the idea that textbook for active influence should be created also we see some connection that can be applied to organizational management for creating translations to make it multi-language from the beginning and also we should see for some domain specific use cases that we can understand in terms of wet ontology what we are going to create thanks alex so on to the questions section we're going to start off pretty broad here in the edu how do we go about determining the core ideas and terms for active inference this will be the format of the question slides carl so feel free to jump in right i guess um it will be structured around the um the key ideas um and essentially ingredients that underwrite the um the free energy principle and how that translates into active inference so you know without thinking about it too deeply my mind just goes to what are the things what are the basic ingredients you need to explain to somebody and what active inference is and and why it works and it normally starts off with the notion of a generative model and then from that um you spin off um all the appropriate um mathematical ideas and constructs and descriptions that would would attend that um i mean it may be best to um reflect the question back to you so this is a really neat idea having an ontology and it's certainly my experience that people are entertained by um sometimes the poetic use of phrases and descriptions such as um epistemic reforms when trying to grapple with you know what what are the fundamental ideas behind active inference some of them are

fundamental and some of them are not so it's so it's certainly an interesting idea to try and tie down the ontology but let me let me ask you just means what it says in the sense that you're trying to define the essential concepts and how they relate to each other is that is that the basic idea yep going back to this slide here we want to have a continuum from a list of terms potentially that could be developed into coherent and again principled course material and competencies but also develop a logic and we're developing within the sumo ontology development framework which defines not just relational edges but actually uh actual logic and so we hope to be able to ask like is this a complete active inference model have we really checked off all the boxes and used those kinds of logical tools that are accessible to the well-developed ontological frameworks okay well that's very compelling and um very clear um it strikes me then that you you know it would be useful to link that operational ontology to the underlying maths so you're much of the much of the conceptual steps both in understanding and implementing um active inference usually in terms of simulating your interesting behavior or um using it as an observation model to explain some empirical uh data from from a study much of it um can be developed in terms of a series of moves um that usually or in fact almost universally are inherit from the are framed in terms of [Music] either information theory or linear algebra or differential equations and you can just build the story from that so um if you're looking for that degree of formal and useful um uh detail um then it would be one principle you might refer to is basically what your where does one equality assertion or description or variable or object where does it come from in terms of inheriting from the the more basic formula so what i'm thinking of here is where does active inference start and how do you get to the calculus and the bayesian mechanics that you would you you'd associate with with active inference and my guess is given the structure or the way that you have uh approached the ontology you've probably actually done that already or are in the process of doing that are you going to go through some examples that would sort of highlight the you know the strategy and the um and the problems which are usually more illuminating than the solutions that you've encountered so far sure i'll switch here to this screenshot the the current state of what it looks like and we're starting just in tabular form by compiling up to five references and citable definitions first just looking for exact cases where a term is used and then we'll go from how the term has been used towards synthetic definitions that capture different senses of the term and then along with the concise narrative of the field and also ontology experts who are here with us we're going to then be working to make the actual logical underpinnings elucidated in terms of specifiable code rather than just concise english

definitions and then from that sort of generator of the formal relationships will be able to descend into mathematical formalisms or other natural human languages yeah well we'll keep you posted on this project though for sure let's go to this next question and imagine that we had that set of terms in development it's going to be a work in progress our whole lives how would we go from core terms and ideas to an interactive and enlivening education that speaks to people from many different backgrounds so i i'm going to answer this question from the point of view of my experience as a supervisor which is probably a little bit of narrow of a narrow remit from your more general ambition uh and i imagine that this is uh related to this notion of was it sort of debt i can't remember now but this notion that you don't want to put too much pressure on people um when becoming acquainted with the utility and application of um of either the code or all the ideas so um in my experience with um in an academic setting uh just having toy simulations is usually the best way to give people a feel for what these uh what this approach does and how it can be used so um it's you're enormously potent in terms of demystifying and also illustrating the functionality at hand or that can be accessed um it also having a sort of um you know a working um working or at least a toy model um sort of provides a proof of principle and that can um strip away the magic as well and i think your your ambition to try and make this accessible to people who are not necessarily fluent in the underlying information theory or dynamical systems it is is very laudable and perfectly feasible so again in my experience some of the most creative applications of active are can be by people who don't really necessarily wonder too much what's underneath the hood it all comes back again to the design of the generative model so if you get the if you get the generative model right um and it's apt to describe the thing that you want to understand or to simulate um then usually everything else follows uh follows suit and i mean that in the sense that you can just take off the shelf software which i presume that you your ultimate ambition is to make available um and make it work in the service of so say well what would this agent or this synthetic creature or person um do in exchange with um her environment if this was the generative model and this was the genetic process so a lot of this um a lot of this really i imagine in terms of answering your question how do we go from core terms to interactive and enlightening is just establishing a language a lexicon that allows you to talk through somebody in constructing their own um simulations that speak to the issues that engage them um either academically or be uh or beyond academia um so um clearly then the core terms play the role of uh literally a language in terms of communication which brings us back again to the you know the importance of the ontology um and having the terms um linked in a formal way to

the uh the the mathematical expressions and also procedures and processes um you know so i guess that a precondition to do the uh to use the core terms in an interactive and enlivening educative sense will rest upon getting that and getting that ontology right in my experience you know the best way to get the ontology right in the sense of it being enabling um is just to talk about the terms and until everybody until there's some consensus and everybody understands them in terms of either their terminology well so i know in both in terms of their technology but also in terms of where they come from from the point of view of the code and ultimately the maths underwrites all this is that the sort of um answer you're looking for here or thinking along with are you thinking along the same lines it sounds great there's so many dimensions there just to provide a summary or just jump in at one place what is active inference and what does active inference do right so that's perfect because i was just thinking it would be really useful just to go down the sort of the the terms that you had in the um in the um previous but one slide highlighted in green because i think you know all the heavy lifting here is is really just shouting about um what what are the core um aspects and claims or um the thing the core things that you're trying to communicate with any one of those terms so for me active would be a description of a process that can be seen as something that arises from the free energy principle so you can either tell that story from the point of view of a physicist and say that active inference is a technological description of processes that systems that self-organize must possess or you could tell the story or define active inference from the point of view of neurobiology and histology um you know from a point of view say predictive processing and describes what what it entails um and i've used the word basic mechanics before because um from the point of view you know the physics definition it would be um a teleological description of a bayesian mechanics that necessarily arises um with some certain with certain assumptions from any self-organizing system um one key thing about active inference which um i think would be important to put in the definition in the ontology i'm not sure it's it's already there but you know if you're in charge of sculpting the um then you're in the position to make sure it's there um is it it's about uh it's beyond predictive um processing it's beyond sentience and it emphasizes um or reflects the pragmatic term at the beginning of this century really um sort of epitomized by the sort of the four e's you know embodied um extended and the like um to make it clear that sentience is is active and that you are talking about your um the circular causality of engagement of any um particle personal plant with whatever is out there so that would be certainly one thing um to emphasize in terms of the your what active inference means the inference is interesting in the sense that it does imply a

process and a process with purpose which is to infer which is why i keep using the word the description of something that's actually underneath the hood from the point of view of physics one final point here is um there's an easy confusion i think between first of all active inference and passive inference so that's certainly something which probably needs resolving certainly in the philosophical literature so i often come across um the you know philosophers who say well there's passive influence or perceptual imprints which is just basically inferring uh states of affairs in the world um on the basis of some sensory evidence and then there's the extra bit which is the active bit which is now you're in charge of gathering that sensory evidence upon which you are now going to um prosecute your perceptual that that that's an interesting dichotomy which i'm not sure is you know um is the right is a correct um dichotomy um um if it's not right i'm not sure that it is not right in the sense that it is a useful distinction but certainly is not what active inference was originally um um termed to mean you know by conjoining active and inference there were a number of motivations first it was a generalization of david mccain's active learning um but probably um more importantly it was it was a nod to the notion of active sensing an active and your active perception that perception is in and of itself and an active process a constructive process that you have to put policies plans and action into the game so that i think would be one important um aspect of active inference to define and i don't know that it has been defined so far so you know perhaps it's your job to to define that the other thing which is um important i think in terms of emphasizing what active inference entails actually comes that inactive perspective which is about the consequences of action and that has an important but really simple um concomitant that the consequences of action are in the future and that means you now have to think if you're thinking from about active inference um in terms of of teneology or as a normative um theory of of behavior of sentient behavior and you have to now think about i'm sorry and i should say uh qualify when i say normative i mean it can be operationally defined in terms as an optimization process that in turn requires you to define the objective function or functional and that's important practically because if you're now thinking about sentient behavior active inference and it's influence about things that not haven't yet happened because you haven't yet acted um then you're necessarily talking about objective functions or functionals that are about states of affairs in the future and that is an important move um and something that active inference embraces which goes beyond predictive coding so much of the literature in the 20th year in the 1990s you know and subsequent much of the literature inspired that sort of inactive perception or active sensing take situated cognition um take on um on sentience um originated

in your in things like predictive coding but predictive coding is not what is meant by active inference you can do predictive coding just by um if you're a statistician just minimizing variational free energy um that that that's only um half the game once you move into the world of active inference from a technological perspective all you are you have to do that you have to um form beliefs about hidden states of affairs in the world um using sort of the perceptual side of perceptual inference but that is only in the service of rolling out into the future um and deciding what the best thing is to do next and that running out into the future and deciding clearly calls for an objective function so in active inference that would be the expected free energy which may or may not be unfortunately named but that's that's what it is um and therefore active inference sort of implies that you are committed to optimizing an expected free energy and implicitly it's all about choosing the next thing to do so for me those would be two would be two sort of cardinal things that should be embraced by definition of that you know transcend other normative approaches so for example you know reinforcement learning in behavioral psychology um would be all about what the good things are to do and you commit to a loss functionality value function of states um your if that was your um the kind of behavior that you're trying to describe if on the other hand you were all about the psychophysics of perception or just building base or digit or terrorist recognition systems where you weren't in charge of gathering those data then your objective functions will be would be very very different um but what active infant says well you can't you can't carve up the two problem domains because they're just both of the same coin and thereby you're you know you're now facing the problem of defining an objective function that is fit for purpose that does both um the belief updating about latent states or hidden states generating the data and also the best way to solicit solicit or cause those data or outcomes under some prior preferences or some goal-directed constraints is that a good long-winded answer thank you for the comprehensive answer it leads directly to our next questions which are what is the free energy principle and especially what is the relationship between active inference and the free energy principle right well that's that's i think a slightly easier question to to to answer and the free energy principle um is um just a variational principle of least action um it um why is it special or not formally identical to all the other variational principles that we use if you look under the hood right from quantum through statistical and stochastic um to classical mechanics well the only thing that differentiates really the variational uh principle of least action that is the uh or that is the free energy principle is that you're paying careful attention to the separation of states to which you apply that principle um the separation of states into the

states of an agent or a particle or a part of a person and the outside states so technically if you were in statistical thermodynamics for example you'd normally assume that separation in terms of some idealized gas that was contained within the container or heat reservoir or a heat bath without really worrying about where the heat bath or the the heat reservoir came from but he says well noah you can't really do that you've really got to attend very carefully to what um license is a separation of different kinds of states so you can assign to the inside of something on the outside of something and the states that mediate the exchange between the inside the outside outside and then you get into the markov blanket and mark off boundary literature so just to summarize a free energy principle is just a a principle of least action by which i mean that there is a description of dynamics in terms of the most likely paths any system will take that is the special provenance of um a partitioning or a separation of the states of some universe into the states that are owned by an agent or a particle and those that are not and the states that mediate the exchange between them so that would be the free energy principle um active imprints as i say is a sort of um technological spin-off from um the the free energy principle in the sense that um in the same sense that you have now at hand a principle of least action um you know that it allows you to identify simulate define the paths or trajectories or the narratives that a system the most likely paths trajectories or narratives that a system will pursue under certain conditions and those conditions are just that there is an attracting set of states um to which that system um will converge to or will look as if it's attracted to um so sorry what i was working towards was the notion of um an attracting set as a metaphor for equipping that physics with a tediology and that tediology is nicely illustrated by the notion of attraction so you know when mathematicians talk about attractors in the uh particular case in the free energy principle these are these sort of pullback protractors or the kind of attractors that you get in random dynamical systems um there's a a proper and natural tendency to think that these particular states um of the attracting set literally attract in the sense of you um gravitational attraction or any other kind of attraction they pull they pull states towards them so that to me would be a technological uh interpretation which i think is much closer to active inference that you're saying that influence is a process that has a purpose and the underlying free energy principle looks uh allows you to say the way it looks as if self-organizing systems uh show these certain properties they're attracted to certain states they show um they're attracted to certain paths and we can describe those in terms of the theological ontology and that would be active inference um one practical difference between active inference and the free energy principle is that the free energy principle is just a principle

it's neither right or wrong it's just like um um millage is uh noted it's like sort of know this theorem or your um or hamilton's principle of um but as soon as you start to say well i think that this principle applies to this population or person or particle and that certainly recommits you or requires you to define a um the the attracting set of states a pullback attractor um in another jargon the equivalent would be a generative model and as soon as you commit to a genetic model to explain the technology of this system or this agent or this person and then you've moved into the world of non-falsifiable principles into falsifiable hypotheses because you could have chosen the wrong charity moral and thereby there will be evidence for choosing this generator model or that generative model so the relationship between active imprints and the free energy principle is operationally quite simple you know active inference is the application of the free energy principle to a particular system but in that application you're bringing a lot of tediology to the table and more specifically um having you're having to commit to a particular generative model and as soon as you do that that becomes your theory or your hypothesis about what is an app description for this system so a number of i think sort of interesting distinctions in terms of the relationship between active inference and free energy that i imagine um your ontology is already addressed or it certainly is certainly addressing well we'll get there thank you for that excellent answer for the next question lorena please read it out oh hi so still in the spirit of broad questions and broad terms uh and that i think comes in line came before so how and where does the idea of information play a role in the free energy principle and how does it relate or is active in the sense what is something to keep in mind when thinking about information dynamics in active inference right bill these are great questions i'm getting the hang of this now you just want me to talk i've presented a question to another talking which is which i'm very happy to do um are you sure you want me to do that or should this be a conversation perhaps it'll turn into a conversation at some point anyway so information so it plays um a dual role um um in the sense that information theoretic formulations underpin most of the derivations behind that principle of least action and and it can be no other way in the sense that all mechanics from physics is really articulated in terms of probability densities or distributions um and as soon as you um have a mechanics that is um or a calculus or probability distributions you're effectively in the world of information theory and you see that at many different um levels so one nice example of this is that the the central um quantity that we often use to score the likelihood of being in a particular state um if you're a statistician that would be the marginal likelihood if you you know were fluent um with with an fep ontology it would be surprising uh almost simply a surprise and if that is just basically

the self-information it's um if you're a physicist you look at this as a potential it's a negative log probability so you start really um when thinking about the physics with this um central concept of self-information which the iep can be read as a potential or a surprisal function or surprise and it is the thing that the variational free energy is a bound approximation too so that level and then everything else every other move you make mathematically in terms of the expected self-information being the entropy and you know why that is important as a characterization of various probability in the setting of self-organization um and you know would testify to the fact that information theory is absolutely central to all the maths that underlies the the physics of the sentience that emerges from um having a distinction between the states of the system and states that are not in the system namely the markov blanket having said that i think information um to most people's minds um usually means more certainly in the folk psychology context it's really information about something um and the um fep active implants has i think something quite special to the bring to the table here that goes beyond um information theoretic treatments of uh you know that you get in communication and signal processing and rate distortion theorems all of that kind of information is just um your extensions of information theory that inherit from self-information or the implausibility of a particular event or message you know or in your more abstract domains such as sentience and consciousness you would go to something like integrated information theory but that is all about this um shannon-esque kind of information the opposite the other kind of information which is information about something so what i wanted to try and uh put on the table is the very fact that you've got this markov blanket or separation of and states on the inside and states on the outside means that now you can equip the states on the inside with the role of encoding posterior conditional bayesian beliefs about states on the outside and that introduces technically a different kind of information geometry a different kind of information theory where crucially now you can read the internal dynamics as um containing or having information about what's going on on the outside and this is a really important move equipping your neural dynamics or variational message passing or belief propagation in a computer with an information geometry that now allows you to read off the state of the computer or the state of the neural activity in terms of what it is encoding or belief or the information it contains about the outside and so that that sort dual aspect information geometry has been celebrated to a minor extent in the philosophy literature by vanya um vice um you know asking the question is you know is this really the maths of sentience where you now have information about things and in a sense that you know that really is the heart of the free energy

principle in the sense of active inference anyway in the sense that it equips um that jewel that information geometry i mean technically what you are saying is that any particular internal state of a computer or a person or a brain now can be read as encoding a bayesian or a posterior belief about um other states namely hidden or latent causes outside the markov blanket and that technically something called a statistical manifold and as soon as there is a statistical manifold there's an information geometry and any movement on that manifold necessarily implies a change in your bayesian beliefs namely bayesian belief updating which means now there's a an interpretation of neuronal dynamics movements on a statistical manifold on the inside in terms of belief updating so the the notion of active inference as the process of belief updating really um you know rests upon this fundamental notion that there's information about stuff going on um about stuff that is encoded or parameterized by the by the internal machinations and the mechanics and the dynamics of of so i think it might be quite important to if you're trying to describe or educate um people in terms of how they should understand information as playing a central role in sentience i think it'd be important to differentiate the um the mathematical notions of shannon information and self-information and the countless of probability and mutual information for example and say that this is not the kind of information that is implicit in an information geometry and the sentence that is afforded by active inference when now you're understanding neuronal dynamics message passing in the computer on on some um on some sort of factor graph and uh because in this instance each of those messages or that those neural dynamics um can now be read as belief updating namely changing your mind about other things so that the stuff on the inside has about stuff on the outside thanks for this important answer and we're going to pass over a few questions and go to 18 with stephen reading the question to continue on this theme about the separation of the inside and the outside so thank you stephen and please read off 18. thank you um it's going to ask how can the integrity of the active inference process theory be maintained when blankets blanket states and generative models are being interpreted in novel ways so we were thinking about um what do you think of discussions around markov or pearl fristen blankets etc all right that's an excellent question i mean i have a quite a technical answer um so if it's getting too technical tell me i'll try and get back to what you know um what you what you um we're really trying to unearth um so this is not a fast moving field but certainly been a a delicate and important um area of discussion over the past few years so in the original introduction of markov blankets there was um an explicit nod to um pearl's construction of markov blankets and how rock of blankets are used practically in terms of simplifying

message passing in in computer science however that may have been um something of an oversimplification and because the from the point of view of the free energy principle um it is um the the kind of causality that the uh the free energy principle deals with is not the kind of causality that um people particularly people like pearl but also people dealing with things like granger causality um deal with um so we're not from the point of view of the free energy principle that starts with um a stochastic differential equation or a random dynamical system written as a random dynamical equation and our processes being simple examples uh in physics these would be like equations common to all of these starting points is time and evolution and dynamics now there is nothing in pearl's formulations well certainly there's nothing in pearl's book on causality that deals with time um and i know that because i uh before the days of pdfs and being able to go and search particular words i had to go through and find out there's one paragraph that mentions dynamics um so you know if you were in in statistics computer science you know these this will be the world of dynamic bayesian nets um the the the is their take on something which is actually much more universal which is basically um the universe as a markovian dynamical process so just stepping back the the challenge now is to articulate um independence independences that underwrite markov blankets in the sense of pearl in terms of dynamics so you've now got link two quite distinct fields which is basically the fields of dynamics and launch van processes and things that have paths of least action to the world of statistics and pearl-esque independences and causality as interventions that have observable consequences the problem in doing that linking is that you have to um really abandon the notion of causality in the world of range of causality in pearl because causality is baked into an inherent in writing down any um differential equation be it stochastic or random or deterministic in the sense that states cause motion so the causality in this context would be a more control theoretic causality so that means that you can't then use the causality concept later on but it does mean that you've now got to derive from a dynamical markovian calculus um the um the necessary um conditions that would lead to um the conditional independences that are necessary to define markov boundaries um just to slip in here um the markov blanket um is um if you is composed of minimal blankets mainly boundaries in the sense of pearl and on most recent analyses it it looks as if the blanket is actually two markov boundaries um in the sense of pearl but to get to the sense of pearl you've got to think very carefully about what are the constraints that lead to the the conditional independences where those constraints are specified in terms of equations of motion um and um you know and things like the amplitude of random fluctuations so once you've um once

you've seen that that is the link that needs to be made that actually um simplifies the thing it simplifies things in the sense there's no real attitude for interpretation so i'm going back to your the part of your question you know blankets and genitive models are being interpreted in novel ways and i i don't think there's any latitude any novel interpretation other than the sorry if in novel ways you mean the best way or the correct way and we just haven't got that yet then i would you know i'd concur entirely with that if you think that there is some latitude there's some library of uh insightful reinterpretations and redefinitions uh all of which have equal veracity then i i would suggest that's not the case there's only one way and there's only one markov um or there's only one particular partition um that will give you that can be articulated in terms of markov blankets and the only novelty there is really in tying down very precisely and um and defensibly how you get from a longevan formulation to a markov blanket um at the moment the novel way of doing that um looks as if it's a that the conditional independence is a rise from sparse dynamical coupling of causal coupling so if you read um the the causality as the influence that a state has on the motion of itself or any other state in this sort of minimal launch run like uh description of the universe um then it is the sparsity of influence on the sparsity of coupling that leads to conditional independences and if the system has a sufficiently rich sparsity of conditional independences and implicitly coupling then it will have a particular petition and if it has that particular petition then the then the free energy principle holds so i think the discussions around markov pearl fristen and blankets are essential they're fascinating and the conclusions of those discussions that i think are gonna probably um have to refer back to the underlying maths and that maths is all about connecting longer formulations of physics to the kind of calculus that pearl um has established um you know in a more statistical sense thank you for the educational answer this brings us almost to the end of the dot edu section so i will pass to the final question to be read by dean who had several excellent points and questions so dean feel free to ask however you would like uh good morning uh so the question is what's the difference between a subject matter expert and a prediction prediction matter expert and how does this relate to your mode of interaction you're going to have to unpack what subject and prediction matter experts means for me yeah so for me interesting you become a subject matter expert by gaining a certain amount of concentration in a in a particular field or area and you become a prediction matter expert when you are able to think more distributively more dispersively and so i think what when i read some of the things and listen to some of the stuff that i've heard you talk about you you've brought these two worlds together and so i'm kind of interested in hearing what you think in

terms of introducing some of the the ideas and principles that you brought into a world where traditionally we focused on concentrating whether it's materializing something from an engineering perspective or deciding what's in and what's out you've brought in another aspect to look at and i'm i'm just curious what you think of that okay that's a fascinating um distinction i'm not sure it's terribly important when i think about it because clearly you're the expert on this but it certainly would be fascinating to consider the conditions under which you were able to simulate the emergence of a subject matter versus a prediction matter expert in silicon for example just as a proof of principle that these are both um effectively based optimal ways of responding to a particular environment and my guess is that you you would be able to do that relatively easily by appealing to the ideas that you find applying some of active inference notions to structure learning and development where the basic idea is if you've got a very volatile environment by which i mean that there are there's lots of uncertainty in the contingencies or possibly there are lots of random fluctuations um that are irreducible in terms of your your ability to to predict the the outcome of the trajectory of latent states of the world in which you are becoming an expert then you um when you parameterize your uncertainty you're usually formally in terms of the precision of various likelihood mappings or probability transition um matrices in a sort of discrete state space generative models when you parameterize your beliefs about that uncertainty um irreducible and volatility then agents that believe or have inferred that they are in a very volatile changeable capricious world usually become better at the prediction side of things in the sense that they um rely less upon um deep past experience and assign more precision or um more potency to um the more recent evidence so they have a different style of evidence accumulation that enables and also they have the level of uncertainty about what will happen next so it looks as if in their predictive engagement um and epistemic foraging in that world it looks as if they are better at predicting changes because they're not committed to a particular uh a particular um explanation or understanding of how their world works on the other hand if you create a world which is incredibly predictable and learnable then over time the natural pressure to minimize um free energy um translates into a pressure to minimize complexity namely a way of modeling your world and your exchange with it in the simplest way possible in accordance with oklahoma's principle and what that lens leads to is somebody it sounds as if it's somebody who becomes a subject matter expert but so the subject matter is their lived world that has now become so predictable that they do not entertain all possible other outcomes because they have precise beliefs about the way that things will unfold and they can make very wise very parsimonious

or using parsimonious degrees of freedom they can make um they can make moves and become very expert in the way that this particular non-volatile predictable i.e precise world works and the link with aging here if you allow for the fact that we create our own environments and your many levels of active inference will um permit or is a way of framing our economic construction what tends to you know what what people the story people tend to tell is that as you get older you basically make your world more predictable and you become a subject matter expert in your own lived world so i like the example i i no longer go bungee jumping i'll go to discos because my world is very very predictable and i you know very much an expertise because my world is basically my conservatory in my study and my bedroom i i'm a complete ex subject expert on you take me out on you know to disco and i will not i will not be able to predict what's going to happen next because i'm old whereas you know adolescents and children and certainly and you know newborn infants or newborn artifacts discovering their world which is full of uncertainty and they are not yet subject experts and and the epistemic pressures or motivation for them to learn about you know what happens if i do that and what can i control what can't i control that will make them very quickly into prediction experts until they become until they become sufficiently fluent that they can now um engineer their world to make it non-volatile and then they presumably will become subject matter experts so yeah i'm sure that would be that would be fairly simple to simulate using you know all the toy active inference schemes that we currently currently use and it would be really interesting if these two different kinds of synthetic agents did develop some cognitive styles and confidence in what they were doing that looked exactly like the distinction you're talking about i'm not sure it would work but if it does that would be you know i think an illuminating proof of principle thanks for this answer and for this session from the lab and edu that last answer really spoke to the importance also of intergenerational learning at this point we will take a five minute break and we will return for dot coms so thanks again and we'll see everybody in five minutes right here